Zuobai Zhang

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Experience

NVIDIA Research — **Santa Clara, California, USA**
Research Scientist — *May. 2026 – Present*

- Contributed to the development of the Proteina deep generative model suite for protein structure design.
- Proposed a generative protein binder design model Complexa, achieving successful wet-lab validation.

Education

Mila - Québec AI Institute / Université de Montréal — **Montréal, Québec, Canada**
Ph.D. in Computer Science — *Sept. 2021 – Jun. 2026 (expected)*

- Advisor: Jian Tang
- Research Direction: Graph Representation Learning, Drug Discovery, Protein Science

Fudan University — **Shanghai, China**
B.S. in Computer Science and Technology (Honor Class) — *Sept. 2017 – Jun. 2021*

- Advisor: Zhongzhi Zhang
- Research Direction: Graph Mining, Network Science, Spectral Graph Theory

Internship

NVIDIA Research — **Santa Clara, California, USA**
Research Scientist Intern — *July 2024 – Dec. 2024, June 2025 – Oct. 2025*

- Worked on fundamental protein structure generative models with Karsten Kreis and Arash Vahdat.

ByteDance Inc. — **Shanghai, China**
Machine Learning Engineering Intern — *Sept. 2020 – Jan. 2021*

- Worked in Data-Rec Shanghai Group led by Yitan Li.
- Designed machine learning algorithms to improve the recommender system of Toutiao and developed new features in real-world products.

Publications

(* denotes equal contribution)

- **Scaling Atomistic Protein Binder Design with Generative Pretraining and Test-Time Compute**
  Kieran Didi*, **Zuobai Zhang***, Guoqing Zhou*, Danny Reidenbach*, Zhonglin Cao*, Sooyoung Cha*, Tomas Geffner, Christian Dallago, Jian Tang, Michael M. Bronstein, Martin Steinegger, Emine Kucukbenli, Arash Vahdat, Karsten Kreis
  ICLR 2026

- **Towards All-Atom Foundation Models for Biomolecular Binding Affinity Prediction**
  Liang Shi*, **Zuobai Zhang***, Huiyu Cai, Santiago Miret, Zhi Yang, Jian Tang
  ICLR 2026

- **La-Proteiná: Atomistic Protein Generation via Partially Latent Flow Matching**
  Tomas Geffner*, Kieran Didi*, Zhonglin Cao, Danny Reidenbach, **Zuobai Zhang**, Christian Dallago, Emine Kucukbenli, Karsten Kreis, Arash Vahdat
  ICLR 2026

- **Fast Proteome-Scale Protein Interaction Retrieval via Residue-Level Factorization**
  Jianan Zhao, Zhihao Zhan, Narendra Chaudhary, Xinyu Yuan, **Zuobai Zhang**, Qian Cong, Jian Zhou, Sanchit Misra, Jian Tang
  ICLR 2026

- **Proteiná: Scaling Flow-based Protein Structure Generative Models.**
  Tomas Geffner*, Kieran Didi*, **Zuobai Zhang***, Danny Reidenbach, Zhonglin Cao, Jason Yim, Mario Geiger, Christian Dallago, Emine Kucukbenli, Arash Vahdat, Karsten Kreis*
  ICLR 2025 (*oral*)

- **Bering: joint cell segmentation and annotation for spatial transcriptomics with transferred graph embeddings.**
  Kang Jin*, **Zuobai Zhang***, Ke Zhang, Francesca Viggiani, Claire Callahan, Jian Tang, Bruce J Aronow, Jian Shu
  Nature Communications 2025

- **Multi-Scale Representation Learning for Protein Fitness Prediction.**
  Zuobai Zhang*, Pascal Notin*, Yining Huang, Aurelie Lozano, Vijil Chenthamarakshan, Debora Susan Marks, Payel Das, Jian Tang
  NeurIPS 2024

- **Cell-ontology guided transcriptome foundation model.**
  Xinyu Yuan, Zhihao Zhan, **Zuobai Zhang**, Manqi Zhou, Jianan Zhao, Boyu Han, Yue Li, Jian Tang
  NeurIPS 2024

- **Pretrainable Geometric Graph Neural Network for Antibody Affinity Maturation.**
  Huiyu Cai*, **Zuobai Zhang***, Mingkai Wang*, Bozitao Zhong*, Yanling Wu, Tianlei Ying, Jian Tang
  Nature Communications 2024

- **Evaluating Representation Learning on the Protein Structure Universe.**
  Arian R Jamasb*, Alex Morehead*, Chaitanya K Joshi*, **Zuobai Zhang***, Kieran Didi, Simon V Mathis, Charles Harris, Jian Tang, Jianlin Cheng, Pietro Liò, Tom L Blundell
  ICLR 2024

- **Str2Str: A Score-based Framework for Zero-shot Protein Conformation Sampling.**
  Jiarui Lu, Bozitao Zhong, **Zuobai Zhang**, Jian Tang
  ICLR 2024

- **DiffPack: A Torsional Diffusion Model for Autoregressive Protein Side-Chain Packing.**
  Yangtian Zhang*, **Zuobai Zhang***, Bozitao Zhong, Sanchit Misra, Jian Tang
  NeurIPS 2023

- **Pre-Training Protein Encoder via Siamese Sequence-Structure Diffusion Trajectory Prediction.**
  Zuobai Zhang*, Minghao Xu*, Aurélie Lozano, Vijil Chenthamarakshan, Payel Das, Jian Tang
  NeurIPS 2023 (*spotlight*)

- **FusionRetro: Molecule Representation Fusion via In-context Reactions for Retrosynthetic Planning**
  Songtao Liu, Zhengkai Tu*, Minkai Xu*, **Zuobai Zhang***, Peilin Zhao, Jian Tang, Rex Ying, Lu Lin, Dinghao Wu
  ICML 2023

- **Protein Representation Learning by Geometric Structure Pretraining.**
  Zuobai Zhang, Minghao Xu, Arian Jamasb, Vijil Chenthamarakshan, Aurelie Lozano, Payel Das, Jian Tang
  ICLR 2023

- **PEER: A Comprehensive and Multi-Task Benchmark for Protein Sequence Understanding.**
  Minghao Xu*, **Zuobai Zhang***, Jiarui Lu, Zhaocheng Zhu, Yangtian Zhang, Chang Ma, Runcheng Liu, Jian Tang
  NeurIPS 2022 Datasets and Benchmarks Track

- **Effects of Stubbornness on Opinion Dynamics.**
  Wanyue Xu, Liwang Zhu, Jiale Guan, **Zuobai Zhang**, Zhongzhi Zhang
  CIKM 2022

- **Neural-Symbolic Models for Logical Queries on Knowledge Graphs.**
  Zhaocheng Zhu, Mikhail Galkin, **Zuobai Zhang**, Jian Tang
  ICML 2022

- **Structured Multi-task Learning for Molecular Property Prediction.**
  Shengchao Liu, Meng Qu, **Zuobai Zhang**, Huiyu Cai, Jian Tang
  AISTATS 2022, NeurIPS AI for Science Workshop 2021

- **Biharmonic Distance-Based Performance Metric for Second-Order Noisy Consensus Networks.**
  Yuhao Yi, Bingjia Yang, **Zuobai Zhang**, Wanyue Xu, Zhongzhi Zhang, Stacy Patterson
  IEEE Transactions on Information Theory

- **Neural Bellman-Ford Networks: A General Graph Neural Network Framework for Link Prediction.**
  Zhaocheng Zhu, **Zuobai Zhang**, Louis-Pascal Xhonneux, Jian Tang
  NeurIPS 2021

- **Learning Interpretable Cellular and Gene Signature Embeddings from Single-Cell Transcriptomic Data.**
  Yifan Zhao*, Huiyu Cai*, **Zuobai Zhang**, Jian Tang, Yue Li
  Nature Communications

- **Minimizing Spectral Radius of Non-Backtracking Matrix by Edge Removal.**
  Zuobai Zhang, Zhongzhi Zhang, Guanrong Chen
  CIKM 2021

- **Coherence Scaling of Noisy Second-Order Scale-Free Consensus Networks.**
  Wanyue Xu, Bin Wu, **Zuobai Zhang**, Zhongzhi Zhang, Haibin Kan, Guanrong Chen
  IEEE Transactions on Cybernetics

- **Opinion Dynamics Incorporating Higher-Order Interactions.**
  Zuobai Zhang, Wanyue Xu, Zhongzhi Zhang, Guanrong Chen
  ICDM 2020 (short paper)

- **Power-Law Graphs Have Minimal Scaling of Kemeny Constant for Random Walks.**
  Wanyue Xu, Yibin Sheng, **Zuobai Zhang**, Haibin Kan, Zhongzhi Zhang
  The Web Conference 2020

- **Fast Approximation of Coherence for Second-Order Noisy Consensus Networks.**
  Zuobai Zhang, Wanyue Xu, Yuhao Yi, Zhongzhi Zhang
  IEEE Transactions on Cybernetics

- **Nearly Linear Time Algorithm for Mean Hitting Times of Random Walks on a Graph.**
  Zuobai Zhang, Wanyue Xu, Zhongzhi Zhang
  WSDM 2020

Workshop Papers

- **Augmenting Evolutionary Models with Structure-based Retrieval.**
  Yining Huang, **Zuobai Zhang**, Jian Tang, Debora Marks, Pascal Notin
  ICML'24 Workshop ML for Life and Material Science: From Theory to Industry Applications

- **Structure-Informed Protein Language Model.**
  Zuobai Zhang, Jiarui Lu, Vijil Chenthamarakshan, Aurélie Lozano, Payel Das, Jian Tang
  ICLR 2024 GEM Workshop

- **Fusing Neural and Physical: Augment Protein Conformation Sampling with Tractable Simulations.**
  Jiarui Lu, **Zuobai Zhang**, Bozitao Zhong, Chence Shi, Jian Tang
  ICLR 2024 GEM Workshop

- **PDB-Struct: A Comprehensive Benchmark for Structure-based Protein Design.**
  Chuanrui Wang, Bozitao Zhong, **Zuobai Zhang**, Narendra Chaudhary, Sanchit Misra, Jian Tang
  NeurIPS 2023 Workshop on New Frontiers of AI for Drug Discovery and Development

- **Enhancing Protein Language Models with Structure-based Encoder and Pre-Training.**
  Zuobai Zhang, Minghao Xu, Vijil Chenthamarakshan, Aurélie Lozano, Payel Das, Jian Tang
  ICLR 2023 Machine Learning for Drug Discovery Workshop

Preprints

- **Consistent Synthetic Sequences Unlock Structural Diversity in Fully Atomistic De Novo Protein Design**
  Danny Reidenbach*, Zhonglin Cao*, **Zuobai Zhang***, Kieran Didi, Tomas Geffner, Guoqing Zhou, Jian Tang, Christian Dallago, Arash Vahdat, Emine Kucukbenli, Karsten Kreis
- **ProtIR: Iterative Refinement between Retrievers and Predictors for Protein Function Annotation.**
  Zuobai Zhang, Jiarui Lu, Vijil Chenthamarakshan, Aurélie Lozano Lozano, Payel Das, Jian Tang
- **TorchDrug: A Powerful and Flexible Machine Learning Platform for Drug Discovery.**
  Zhaocheng Zhu, Chence Shi, **Zuobai Zhang**, Shengchao Liu, Minghao Xu, Xinyu Yuan, Yangtian Zhang, Junkun Chen, Huiyu Cai, Jiarui Lu, Chang Ma, Runcheng Liu, Louis-Pascal Xhonneux, Meng Qu, Jian Tang
- **A Roadmap for Big Model.**
  I'm a co-first author of this paper.

Selected Projects

- **TorchProtein: A Flexible Machine Learning Library for Protein Science.** (with TorchProtein team)
  A machine learning library for protein science based on TorchDrug, providing representation learning models for both sequences and structures, and fundamental protein tasks like function prediction, structure prediction.
  [Homepage] [Github] [Twitter]
- **TorchDrug: A Powerful Machine Learning Platform for Drug Discovery.** (with TorchDrug team)
  A machine learning platform designed for drug discovery, covering techniques from graph machine learning, deep generative models to reinforcement learning.
  [Homepage] [Github] [Twitter]

Honors & Awards

Outstanding Undergraduates Graduates of Shanghai	**Jun. 2021**
Scholarships for Outstanding Academic Individuals at Fudan University	**Dec. 2020**
Shanghai Scholarship (Top 1%)	**Oct. 2020**
SenseTime Scholarship (29 undergrads from China)	**Jan. 2020**
Chinese National Scholarship (Top 1%)	**Oct. 2019**
Chun Tsung Scholar Program	**Apr. 2019**
Wish Scholarship	**May 2019**
Chinese National Scholarship (Top 1%)	**Oct. 2018**
Gold Medal, ACM-ICPC Asia Regional Contest EC-Final	**Dec. 2017**
Gold Medal, ACM-ICPC Asia Regional Contest Qingdao Site	**Nov. 2017**
Silver Medal, National Olympiad in Informatics, National Finals	**Jul. 2016**
Gold Medal, Winter Camp of National Olympiad in Informatics	**Jan. 2016**
First Prize (2nd place), National Olympiad in Informatics, Shandong Division	**Nov. 2015**

Skills

- **Programming Languages:** Python, C/C++, LaTeX, Julia, JavaScript, MATLAB, HTML
- **Language:** Chinese-Native, English-Fluent (**GRE**: 162+169+4.0)